

User Manual for Spectrum Analysis Web Application

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Version: 1.0 | Date: October 2025

Website: <https://roshanchenglei.github.io/SA>

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1. Introduction

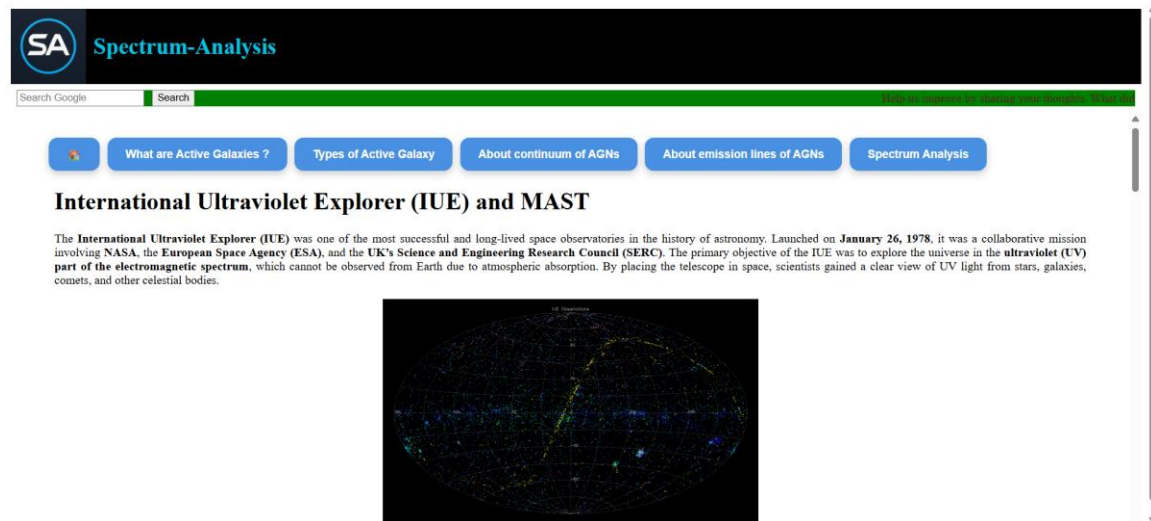
The Spectrum Analysis Web Application is an interactive online platform designed to analyze spectral data of active galaxies. It performs continuum fitting, emission and absorption line analysis, and AGN-based modeling to estimate black hole mass and the size of the broad-line region (BLR). The tool runs entirely in a browser with no installation required.

Supported Browsers: Chrome, Edge, Firefox.

2. Getting Started

1. Open your preferred browser (Chrome, Edge, or Firefox).
2. Navigate to: <https://roshanchenglei.github.io/SA>
3. The homepage will appear as shown below.

Figure 1:



3. Uploading Input Data

The tool accepts plain text (.txt) files containing two columns: wavelength (in Ångström or nm) and flux (arbitrary units). Ensure your file is formatted correctly before uploading.

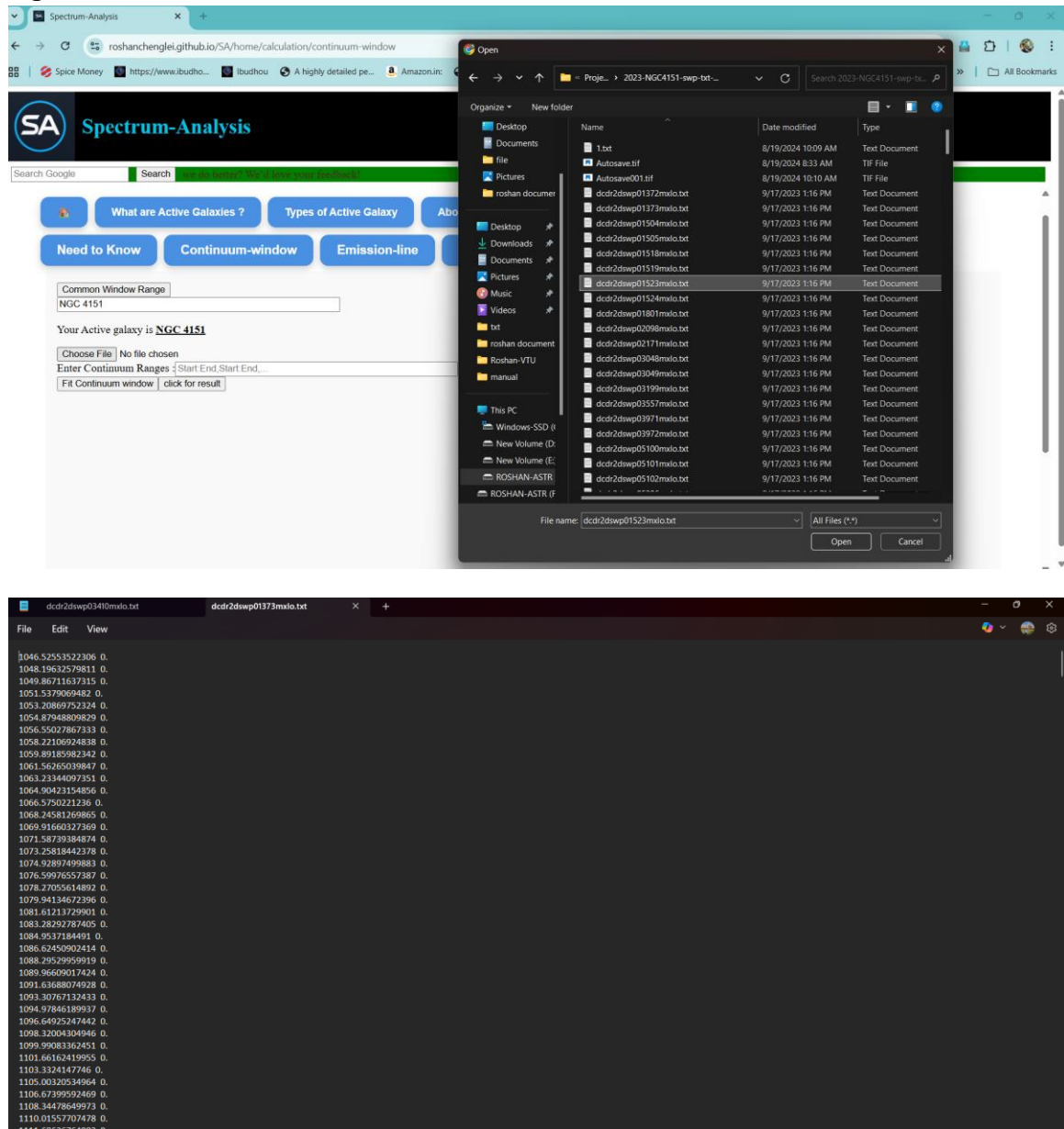
Example Input Format:

Wavelength (Å)	Flux
1300	2.45e-14
1310	2.50e-14

Steps to Upload:

- Click 'Choose File'.
- Select your .txt spectral data file.
- Enter the continuum ranges (e.g., 1200:1250, 1300:1350).
- Click 'Fit Continuum' to check fitting or 'Click for Result' to process.

Figure 2:



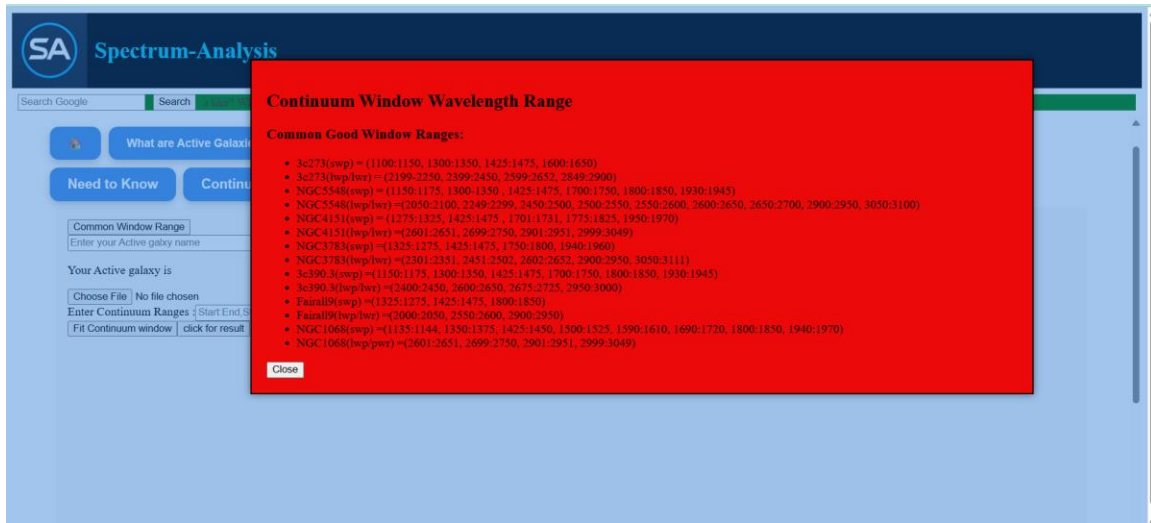
4. Continuum Analysis

This feature fits a power-law continuum to the spectral data within user-defined wavelength ranges. Users can visualize the fitted continuum curve and inspect residuals.

Steps:

- Click on 'Continuum-window'.
- Enter continuum window ranges (e.g., 1300:1350 , 1800:1850,...).
- Click 'Fit Continuum' to perform the fit.

Figure 3:



5. Emission-Line Analysis

This module identifies and models emission features in the spectrum using Gaussian fitting. Users can input multiple emission lines and view parameters such as central wavelength, flux, and FWHM.

Steps:

- Click on 'Emission-line'.
- Enter the emission line name and wavelength.
- If multiple lines, click 'Add Gaussian' for each additional component.
- Click 'Analyze Emission Line' to process.

Figure 4:

Spectrum-Analysis

Search Google

What are Active Galaxies? Types of Active Galaxy About continuum of AGNs About emission lines

Need to Know Continuum-window Emission-line Absorption-line AGN-model

Help Common Window Range

Enter your Active Galaxy name

Your Active Galaxy is

1. Upload Spectrum File

Choose File No file chosen

2. Continuum Fit

Continuum Ranges : Fit Continuum

SA Spectrum-Analysis

Search Google

Enter your emission line Add

Emission Lines:

- C IV Remove

Add Gaussian

Gaussian 1

Amplitude (Actual): 1.3e-12 Amplitude (Min): 1.8e-13 Amplitude (Max): 1.5e-12 Line (Actual): 1548 Line (Min): 1512 Line (Max): 1582 Sigma (Actual): 5 Sigma (Min): 3 Sigma (Max): 7 Min Wavelength (Å): 1512 Max Wavelength (Å): 1582 Wing Window (Å): 5

Analyze Emission Line

Help

Common UV Emission Lines in AGNs:

- Ly α (1216 Å)
- N V (1234 Å)
- Si IV (1393 Å)
- C IV (1548 Å, 1550 Å)
- He II (1640 Å)
- O III] (1663 Å)
- N III] (1750 Å)
- Si III] (1883 Å, 1892 Å)
- C III] (1907 Å, 1909 Å)
- Mg II (2796 Å and 2803 Å)
- [O II] (3727 Å)
- [Ne III] (3869 Å)
- He II (4686 Å)
- [O III] (4959 Å and 5007 Å)
- [N II] (6548 Å and 6583 Å)
- [S II] (6716 Å and 6731 Å)

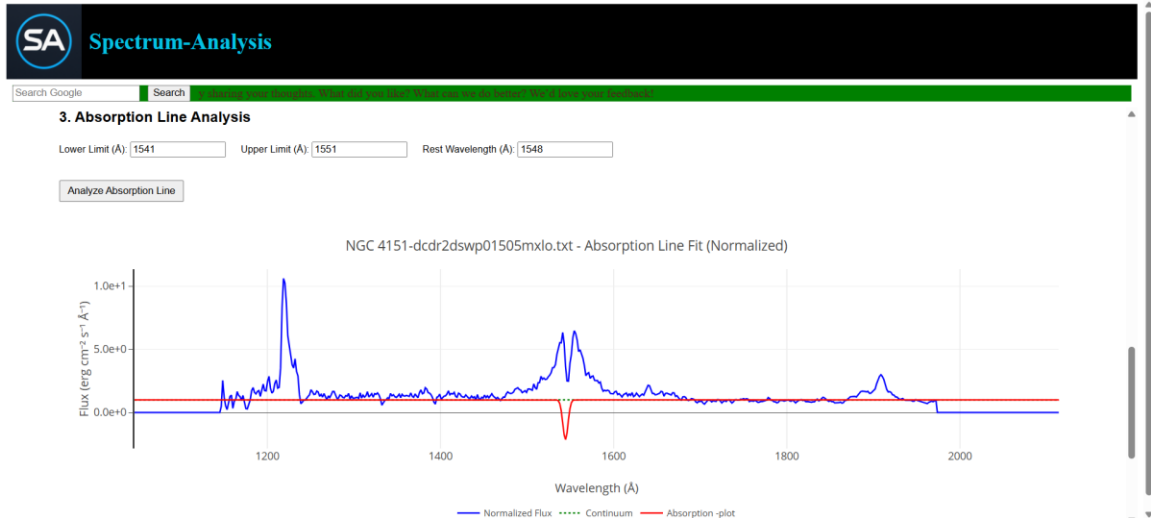
Close

6. Absorption-Line Analysis

The absorption-line module allows users to identify absorption features and estimate equivalent widths or line depths.

Steps:

- Click on 'Absorption-line'.
- Select or enter the expected absorption region.
- Click 'Analyze Absorption Line' to process the data.



7. AGN Model: Black Hole Mass and BLR Size

This feature calculates the black hole mass (M_{BH}) and BLR radius (R_{BLR}) using emission-line parameters and luminosity.

Required Inputs: Luminosity (erg/s) and FWHM (km/s).

Formula Example:

- $$M_{\text{BH}} = f \times (R_{\text{BLR}} \times \text{FWHM}^2 / G)$$

8. Example Workflow

Example: Analyze the galaxy NGC 4151

1. Upload its UV spectrum (.txt file).
2. Perform continuum fitting at 1300:1350 and 2900:2950 ...
3. Fit C IV $\lambda 1549$ emission line.
4. Input line luminosity and FWHM to compute M_{BH} and R_{BLR} .

Figure 7:

The screenshot shows the 'SA Spectrum-Analysis' web application. At the top is a black header with the 'SA' logo and the title 'Spectrum-Analysis'. Below the header is a green search bar with a 'Search Google' button and a 'Search' button. A navigation bar contains five blue buttons: 'Need to Know', 'Continuum-window', 'Emission-line', 'Absorption-line', and 'AGN-model'. The 'Need to Know' button is active. Below it is a 'Help' button and a text input field containing 'NGC 4151'. The text 'Your Active Galaxy is NGC 4151' is displayed. The 'AGN Builder' section includes a 'Spectrum file (.txt — optional):' label, a 'Choose File' button, and a text input field containing 'dadr2dswp01505mxio.txt'. Below this is a 'Select or Enter Emission Line:' label, a dropdown menu showing 'Select emission line', and two text input fields for 'Luminosity (erg/s)' and 'Velocity (km/s)', followed by an 'Add line' button. The 'Emission Lines' section lists: 'CIV — L = 4E+41 erg/s', 'v = 4,300 km/s', 'MBH = 6,380,010.96 Msun', and 'R_BLR = 1.77 ld'. There is a red 'remove' button next to this list. At the bottom are three blue buttons: 'Plot', 'Clear all', and 'Save PNG'.

9. Troubleshooting & Tips

- Ensure the uploaded file is in plain text (.txt) format with two numeric columns.
- If no output appears, refresh the page and try again.
- Check browser console for any data loading errors.
- Use shorter continuum windows if the fit fails.

10. Contact & Support

For support, suggestions, or bug reports, contact:

Khundongbam Roshan Chenglei

Email: roshanchenglei@gmail.com

GitHub: <https://roshanchenglei.github.io/SA>

Formula Used:

1. Continuum model (power-law)

Continuum function

$$F_{\lambda}(\lambda) = A \lambda^{\alpha}$$

• $A \rightarrow$ amplitude

• $\alpha \rightarrow$ spectral_index

2. Continuum fitting (log–log linear regression)

Log transformation

Slope (spectral index)

$$\alpha = \frac{n \sum x_i y_i - \sum x_i \sum y_i}{n \sum x_i^2 - (\sum x_i)^2}$$

Intercept

$$\ln A = \frac{\sum y_i - \alpha \sum x_i}{n}$$

Amplitude

$$A = e^{\ln A}$$

3. Errors in continuum parameters

Variance of residuals

$$\sigma^2 = \frac{1}{n-2} \sum_{i=1}^n [y_i - (\alpha x_i + \ln A)]^2$$

Spectral index error

$$\sigma_{\alpha} = \sqrt{\frac{\sigma^2}{\sum x_i^2 - \frac{(\sum x_i)^2}{n}}}$$

Intercept error

$$\sigma_{\ln A} = \sqrt{\sigma^2 \left(\frac{1}{n} + \frac{(\sum x_i)^2}{n \left(\sum x_i^2 - \frac{(\sum x_i)^2}{n} \right)} \right)}$$

Amplitude error

$$\sigma_A = A \sigma_{\ln A}$$

5. Continuum subtraction

$$F_{sub}(\lambda) = F_{obs}(\lambda) - F_{\lambda}(\lambda)$$

6. Gaussian emission line model

Single Gaussian

$$G(\lambda) = a \exp \left[-\frac{(\lambda - b)^2}{2c^2} \right]$$

- $a \rightarrow$ amplitude
- $b \rightarrow$ central wavelength
- $c \rightarrow$ Gaussian sigma ($\text{\AA}$)

Total model

$$F_{model}(\lambda) = \sum_k G_k(\lambda)$$

7. Emission line flux

Numerical (trapezoidal) integration

$$F = \int_{\lambda_{\min}}^{\lambda_{\max}} G(\lambda) d\lambda \approx \sum_j \frac{G_j + G_{j+1}}{2} (\lambda_{j+1} - \lambda_j)$$

Analytic fallback

$$F = a c \sqrt{2\pi}$$

Absolute flux (IRAF style)

$$F_{abs} = |F|$$

8. Continuum RMS (σ_c)

Normalized residuals

$$r_i = \frac{F_{sub}(\lambda_i)}{\langle F_\lambda \rangle}$$

Sigma-clipped standard deviation

$$\sigma_c = std(r_i)$$

MAD fallback

$$\sigma_c = 1.48 \times median(|r_i|)$$

9. Flux error (IRAF noise formula)

$$\sigma_F = \sqrt{N} \sigma_c \Delta\lambda \langle F_\lambda \rangle$$

- $N \rightarrow$ number of pixels
- $\Delta\lambda \rightarrow$ dispersion (Å/pixel)

10. Equivalent Width (EW)

Definition

$$EW = \int_{\lambda_{\min}}^{\lambda_{\max}} \frac{F_{sub}(\lambda)}{F_\lambda(\lambda)} d\lambda$$

Numerical form

$$EW \approx \sum_j \frac{1}{2} \left(\frac{F_{sub,j}}{F_{\lambda,j}} + \frac{F_{sub,j+1}}{F_{\lambda,j+1}} \right) \Delta\lambda$$

EW error

$$\sigma_{EW} = |EW| \frac{\sigma_F}{|F|}$$

11. FWHM In Ångström

$$FWHM_{\lambda} = 2.3548 \, c$$

In velocity (km s⁻¹)

$$FWHM_v = \frac{FWHM_{\lambda}}{b} \, c_{light}$$

Where,

$$c_{light} = 299\,792.458 \, km \, s^{-1}$$

12. Reduced χ^2 (per Gaussian)

$$\chi^2 = \sum \frac{(F_{sub} - G)^2}{\sigma_{flux}^2}$$

$$\chi_v^2 = \frac{\chi^2}{N - 3}$$

(3parameters: a, b, c)

13. Global reduced χ^2

$$\chi_{v,global}^2 = \frac{1}{N_g} \sum_{k=1}^{N_g} \chi_{v,k}^2$$

14. Wing wavelength (velocity $\rightarrow$ Å)

Half-width

$$\Delta\lambda = \frac{\lambda_0(FWHM/2)}{c_{light}}$$

Blue & red wings

$$\lambda_{blue} = \lambda_0 - \Delta\lambda$$

$$\lambda_{red} = \lambda_0 + \Delta\lambda$$